

Course Learning Objectives (CLO)

The students will be able to:

1. calculate the output to input relation of any physical model in the form of a transfer function using block diagram reduction and signal flow graphs.
2. develop the block diagram of any mechatronic system after analyzing the key inputs, outputs, sensors, transducers and controllers of any physical device.
3. develop the state-space representation of the physical model and analyze the performance and stability of the system in MATLAB environment.
4. interface different sensors, actuators, micro-controllers and data acquisition cards of a given mechatronic device to the computer/laptop.
5. analyze the key features of different type of controllers and develop a suitable controller to obtain the desired performance from the system.

Text Books

1. Bolton, W., Mechatronics: A Multidisciplinary Approach, Pearson Education, New Delhi (2008).
2. Kamm, M.L.J., Mechatronics, Prentice Hall of India, New Delhi (2007).

Reference Books

1. Auslander, D. M. and Kempf, C. J., Mechatronics: Mechanical System Interfacing, Prentice Hall, New Jersey (1996).
2. Neculescu, D., Mechatronics, Pearson Education, New Delhi (2002).
3. Alciatore, D. G. and Hiestand, M. B., Introduction to Mechatronics and Measurement System, McGraw Hill, New Delhi (2005).

Evaluation Scheme

S. No.	Evaluation Elements	Weightage (%)
1.	MST	30
2.	EST	40
3.	Assignments/Lab Work/Project/Quizzes/Class Test etc.	20
4.	Flip Learning Component	10

UME308: Mechanics of Machines

L	T	P	Cr
3	1	2*	4.0

Course Objective: To introduce different types of mechanisms forming different subsystem of machines. To impart the knowledge of vector and matrix methods for position, velocity and acceleration analysis with software tools. To carryout force analysis of engine mechanism analytically. To impart knowledge of force analysis and balancing of rotors. To introduce fundamentals of single degree of freedom vibrating system.

Syllabus

Review of Mechanics: Fundamentals of rigid body mechanics, Newton's laws, D'Alemberts principle, Free body diagrams.

Kinematics of Machines: Introduction to linkages, gears, spur gear, interference in gears, gear trains, screws and cam mechanics, belts, rope, and chain drives as subsystems of machines.

Linkage Mechanisms: Links, kinematic pairs, degree of freedom, inversions, mechanisms, transmission angle and mechanical advantage. Vector and matrix methods for position, velocity and acceleration analysis with relevant software tools.

Friction: Screw friction, clutch plate friction and bearings.

Balancing: Balancing of rotating and reciprocating masses, single cylinder and multi-cylinder inline engines, Field balancing of rotors.

Laboratory Work

Students shall perform experiments based on

1. Centrifugal force
2. Slider Crank mechanism.
3. Cam and follower mechanism.
4. Balancing of rotating and reciprocating masses
5. Gyroscopic effect

Micro Project: Projects for performing position, velocity and acceleration analysis of mechanisms like 4-bar chain, slider crank chain, quick return mechanism etc. to be undertaken which could be correlated to real life situations.

Course Learning Objectives (CLO)

The students will be able to:

1. analyse a set of mechanisms to achieve desired motion transformation.
2. apply analytical methods for the evaluation of velocity and acceleration of mechanisms.
3. analyse and construct cam profiles for different motion of follower.
4. evaluate the unbalance in rotating/reciprocating systems and carry out their balancing.